

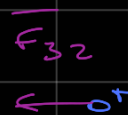
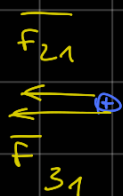
$$\rightarrow \begin{cases} \vec{E}_1 = k \frac{q_1}{\left(l \frac{\sqrt{2}}{2}\right)^2} = \dots & \frac{N}{C} \\ \vec{E}_2 = k \frac{q_2}{\left(l \frac{\sqrt{2}}{2}\right)^2} = \dots & \frac{N}{C} \\ \vec{E}_3 = \dots = \dots & \frac{N}{C} \\ \vec{E}_4 = \dots = \dots & \frac{N}{C} \end{cases}$$

→ stesso segno delle cariche e uguali
→ si annullano a vicenda

$$\rightarrow \vec{E}_{TOT} = \vec{E}_1 + \vec{E}_4 = 108 \frac{kN}{C} \text{ nella direzione di } \vec{E}_1$$

+ - → si sommano!

3) a)



$$\vec{R}_1 = \vec{F}_{21} + \vec{F}_{31} = -k_0 \left(\frac{q_1 q_2}{d^2} + \frac{q_1 q_3}{(2d)^2} \right) \hat{n}_x = -3.15 \cdot 10^{-2} N$$

$$\vec{R}_2 = \vec{F}_{12} + \vec{F}_{32} = \frac{k_0}{d^2} \left(q_1 q_2 + q_3 q_2 \right) \hat{n}_x = -9 \cdot 10^{-3} N$$

$$\vec{R}_3 = \vec{F}_{13} + \vec{F}_{23} = k_0 \left(\frac{q_1 q_3}{(2d)^2} + \frac{q_2 q_3}{d^2} \right) \hat{n}_x = 4.05 \cdot 10^{-2} N$$

b) FORMULA ENERGIA POTENZIALE:

$$\Delta V = \frac{L}{q}$$

$$U = q \cdot \Delta V = -q E d = -k \frac{q_A q_B}{d}$$

- poiché il campo è opposto all'energia potenziale!

